SIMATS ENGINEERING
ASSESSMENT TOOL 1

Course Name: MLA03
Course Code: Reinforcement Learning

COURSE OUTCOME COVERED

CO1: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems. (BL2, BL3)

Assessment Tool Weightage: 50%

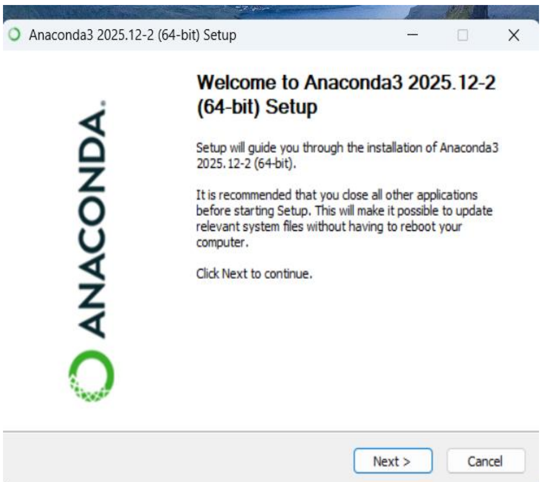
Dr. G. A. Senthil

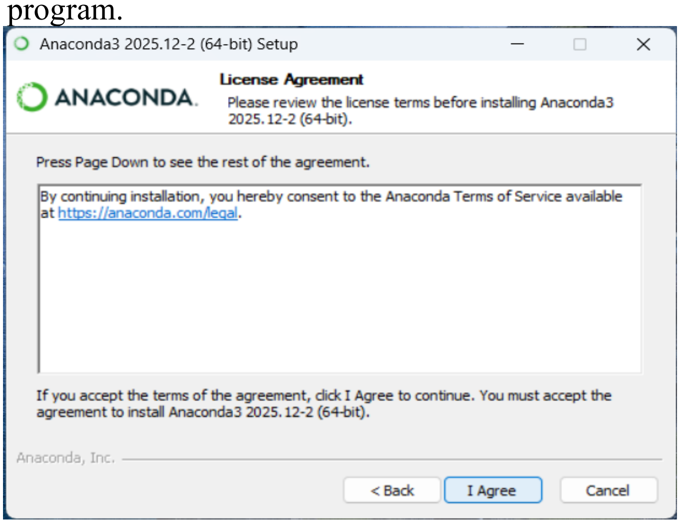
QUESTIONS: 10 **Total Marks: 50**

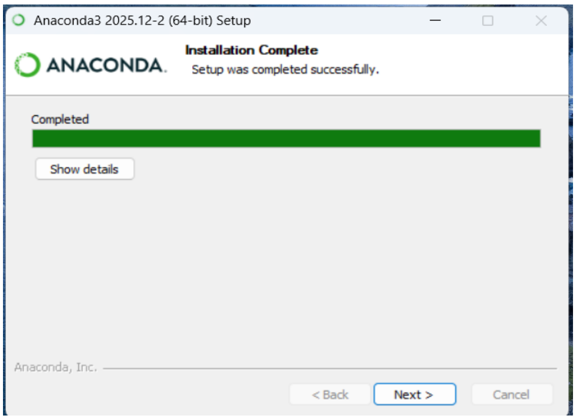
Annexure A

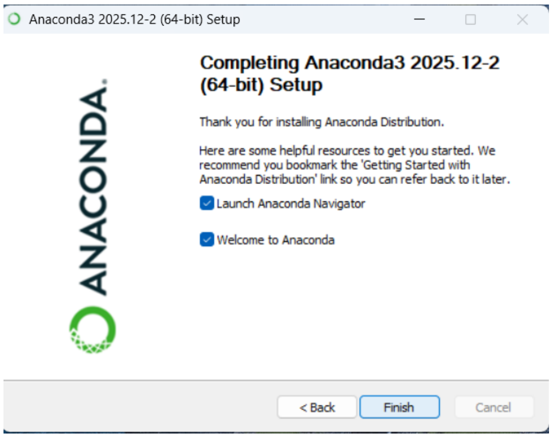
Descriptive Experiment Exam

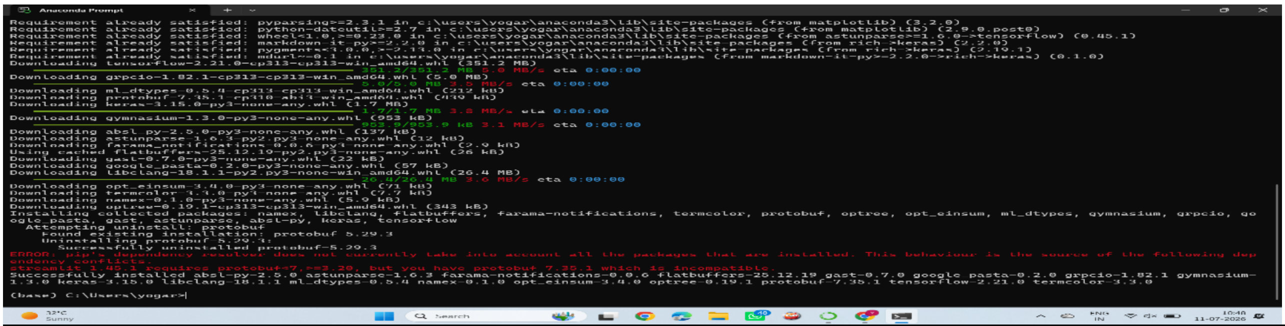
Experiment 1: Installation of Reinforcement Learning Development Environment
Aim: To install and configure Anaconda Navigator, Python, TensorFlow, Keras,Gymnasium (OpenAI Gym), NumPy, Matplotlib, and other required libraries, andverify the environment by executing a simple Reinforcement Learning program.












```
>>> import gymnasium as gym
>>> import numpy as np
>>> import matplotlib.pyplot as plt
>>> import tensorflow as tf
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1783747540.211208   21196 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point r
ound-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1783747545.947677   21196 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point r
ound-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
>>> from tensorflow import keras
>>> print("All libraries installed successfully!")
All libraries installed successfully!
>>> exit()
```

Experiment 2: Familiarization with OpenAI Gym/Gymnasium Environments

Aim: To create, initialize, and interact with standard Reinforcement Learning environments such as FrozenLake, CartPole, and Mountain Car using Gymnasium to understand the concepts of states, actions, rewards, and episodes.

```
(base) C:\Users\yogar>python -m pip show gymnasium
Name: gymnasium
Version: 1.3.0
Summary: A standard API for reinforcement learning and a diverse set of reference environments (formerly Gym).
Home-page: https://farama.org
Author:
Author-email: Farama Foundation <contact@farama.org>
License: MIT License
Location: C:\Users\yogar\anaconda3\Lib\site-packages
Requires: cloudpickle, farama-notifications, numpy, typing-extensions
Required-by:

(base) C:\Users\yogar>exp1.py
'exp1.py' is not recognized as an internal or external command,
operable program or batch file.

(base) C:\Users\yogar>cd C:\Users\yogar\Downloads\RL

(base) C:\Users\yogar\Downloads\RL>dir
 Volume in drive C is Windows-SSD
 Volume Serial Number is 902F-DB84

 Directory of C:\Users\yogar\Downloads\RL

11-07-2026  11:02    <DIR>          .
11-07-2026  10:59    <DIR>          ..
11-07-2026  11:05                60 exp1.py
               1 File(s)             60 bytes
               2 Dir(s)  362,388,631,552 bytes free

(base) C:\Users\yogar\Downloads\RL>python -c "import gymnasium; print('Gymnasium OK')"
Gymnasium OK

(base) C:\Users\yogar\Downloads\RL>python exp2.py
```

```
Environment: FrozenLake-v1
Initial State: 0
Step 1
Action: 0
State: 0
Reward: 0
Step 2
Action: 0
State: 0
Reward: 0
Step 3
Action: 1
State: 1
Reward: 0
Step 4
Action: 1
State: 0
Reward: 0
Step 5
Action: 1
State: 4
Reward: 0
```

```
Environment: CartPole-v1
Initial State: [ 0.04200719 -0.01741832 -0.0238743   0.03254369]
Step 1
Action: 0
State: [ 0.04165882 -0.21218991 -0.02322342  0.31759945]
Reward: 1.0
Step 2
Action: 1
State: [ 0.03741502 -0.01674501 -0.01687144  0.01768408]
Reward: 1.0
Step 3
Action: 0
State: [ 0.03708012 -0.211621   -0.01651775  0.3049965 ]
Reward: 1.0
Step 4
Action: 1
State: [ 0.0328477  -0.0162676  -0.01041782  0.00715037]
Reward: 1.0
Step 5
Action: 0
State: [ 0.03252235 -0.21123861 -0.01027482  0.29652822]
Reward: 1.0
```

```
Environment: MountainCar-v0
Initial State: [-0.5985544  0.       ]
Step 1
Action: 2
State: [-0.5969969   0.00155744]
Reward: -1.0
Step 2
Action: 2
State: [-0.5938934   0.00310349]
Reward: -1.0
Step 3
Action: 2
State: [-0.5892666  0.0046268]
Reward: -1.0
Step 4
Action: 1
State: [-0.5841505   0.00511613]
Reward: -1.0
Step 5
Action: 2
State: [-0.5775827   0.00656777]
Reward: -1.0
```

Experiment 3: Implementation of the Reinforcement Learning Framework
Aim: To implement the basic Reinforcement Learning framework by designing an agent that interacts with an environment through states, actions, rewards, and policies using Python.

```
(base) C:\Users\yogar\Downloads\RL>python exp3.py
Episode Completed
Total Reward: 13.0
```

Experiment 4: Simulation of a Markov Decision Process (MDP)
Aim: To develop a simple Markov Decision Process model and simulate state transitions, reward functions, and policy execution for understanding sequential decision-making problems.

```
(base) C:\Users\yogar\Downloads\RL>python exp4.py
Current State: Start
Action: Move Forward
Next State: Middle
Reward: 5

Current State: Middle
Action: Move Forward
Next State: Goal
Reward: 10

Goal Reached
Total Reward: 15
```

Experiment 5: Bellman Equation Demonstration
Aim: To implement Bellman Expectation and Bellman Optimality Equations for calculating state-value and action-value functions and analyze the convergence of value estimation.

```
(base) C:\Users\yogar\Downloads\RL>python exp5.py
Initial State Values: [0. 0. 0.]
Final State Values:
State 0: 0.810
State 1: 0.900
State 2: 1.000

(base) C:\Users\yogar\Downloads\RL>
```

Code of Conduct:

I Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Yoga madha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understandin g; major misconceptio ns
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand